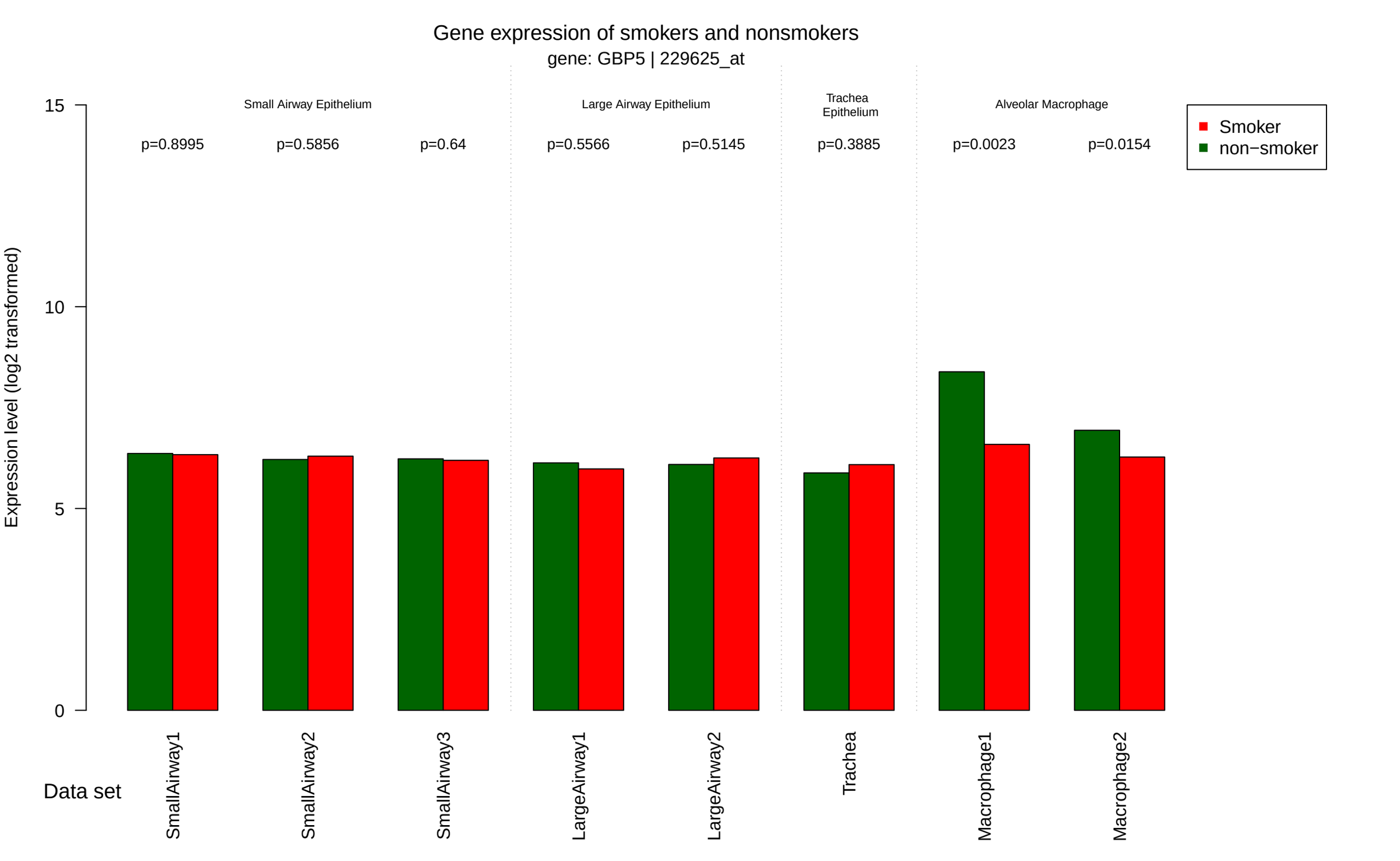
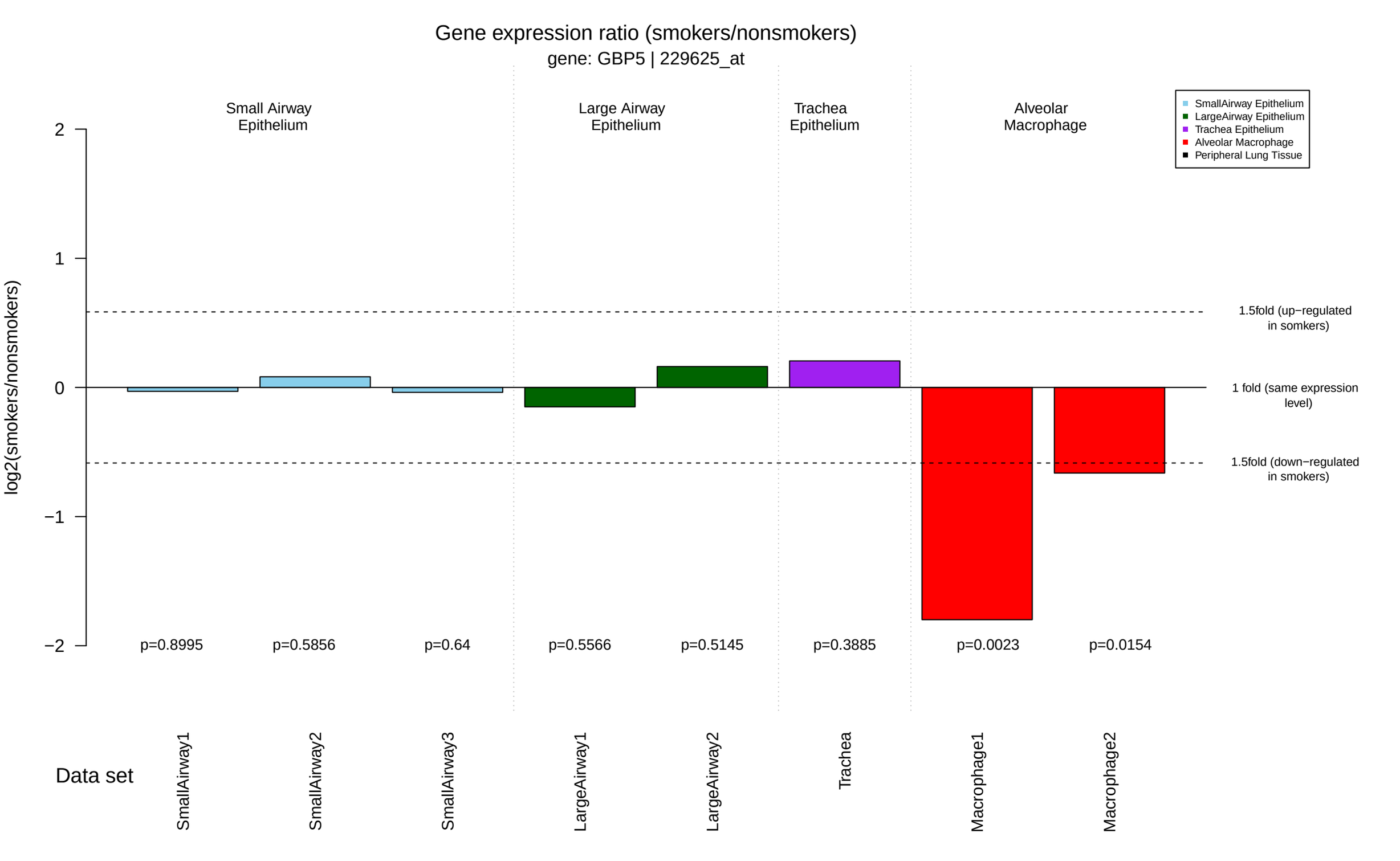
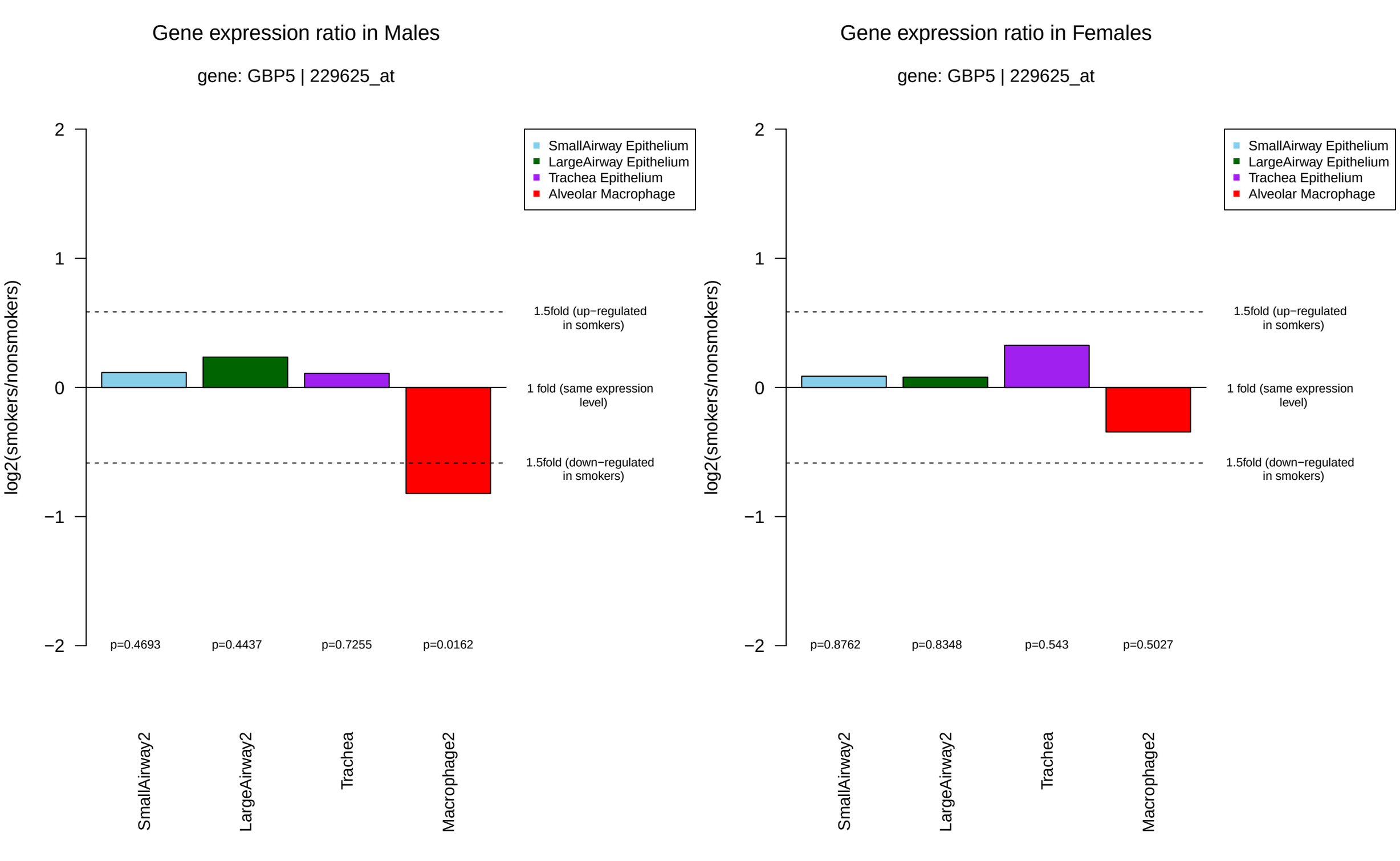
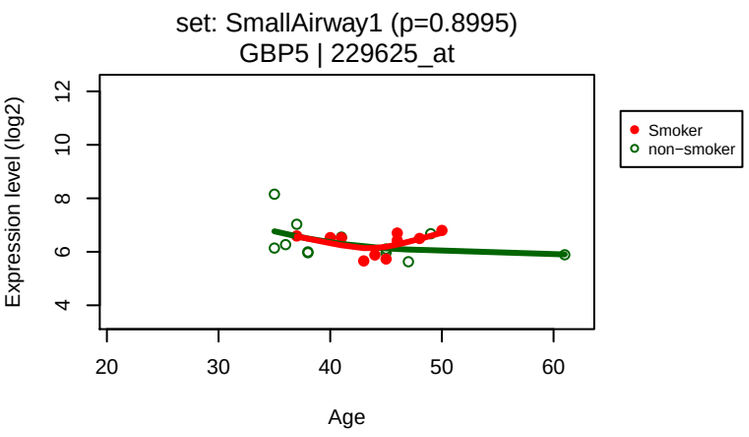
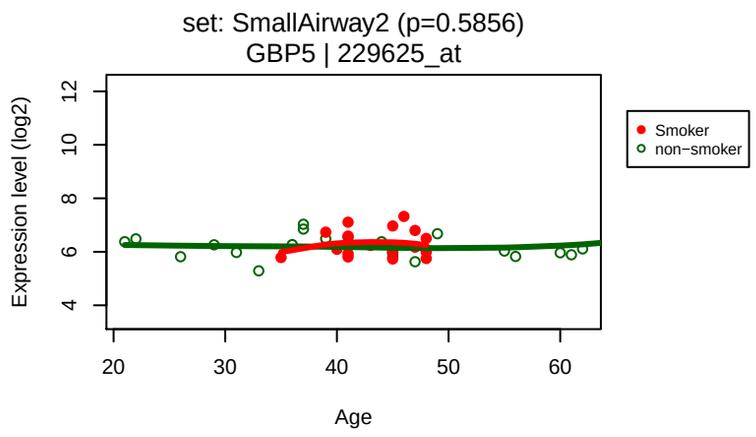
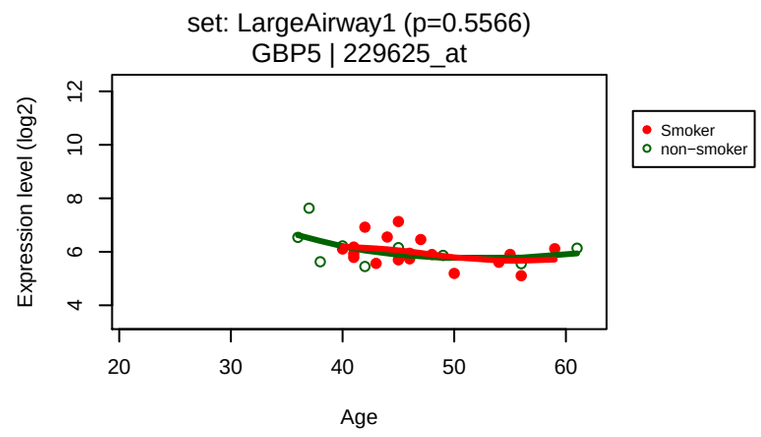

Gene expression of smokers and nonsmokers
gene: GBP5 | 229625_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8995
p=0.5856
p=0.64
p=0.5566
p=0.5145
p=0.3885
p=0.0023
p=0.0154
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

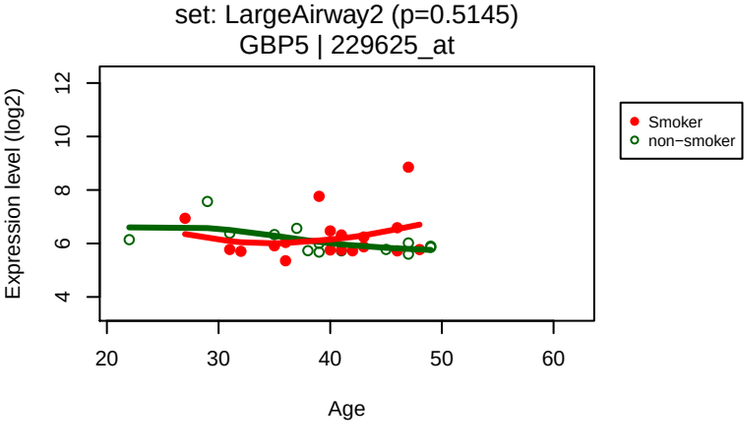

Gene expression ratio (smokers/nonsmokers)
gene: GBP5 | 229625_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8995
p=0.5856
p=0.64
p=0.5566
p=0.5145
p=0.3885
p=0.0023
p=0.0154
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

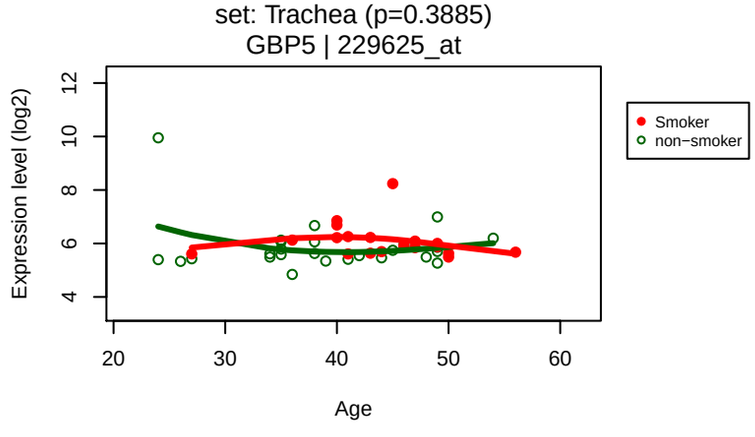

Gene expression ratio in Males
gene: GBP5 | 229625_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4693
p=0.4437
p=0.7255
p=0.0162
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GBP5 | 229625_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8762
p=0.8348
p=0.543
p=0.5027
SmallAirway2
LargeAirway2
Trachea
Macrophage2

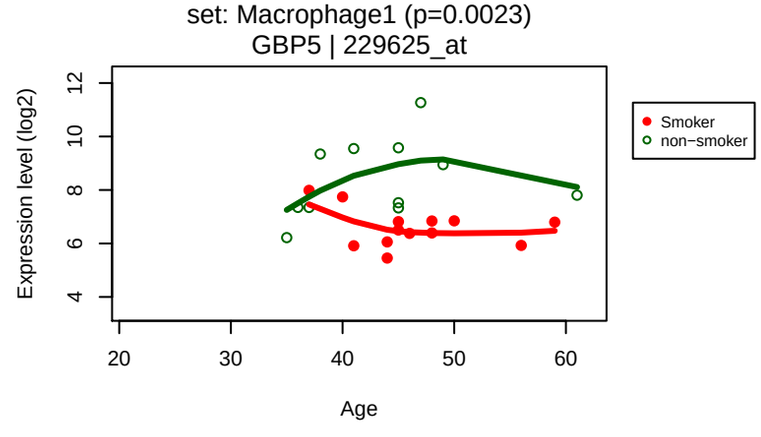

set: SmallAirway1 (p=0.8995)
GBP5 | 229625_at
Expression level (log2)
Age
Smoker
non−smoker

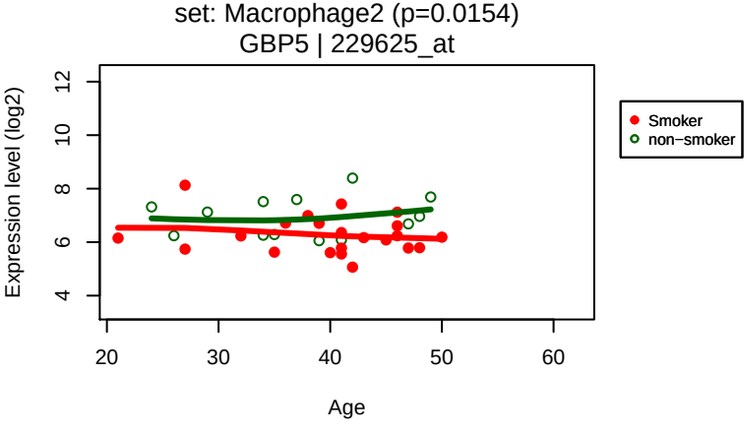

set: SmallAirway2 (p=0.5856)
GBP5 | 229625_at
Expression level (log2)
Age
Smoker
non−smoker

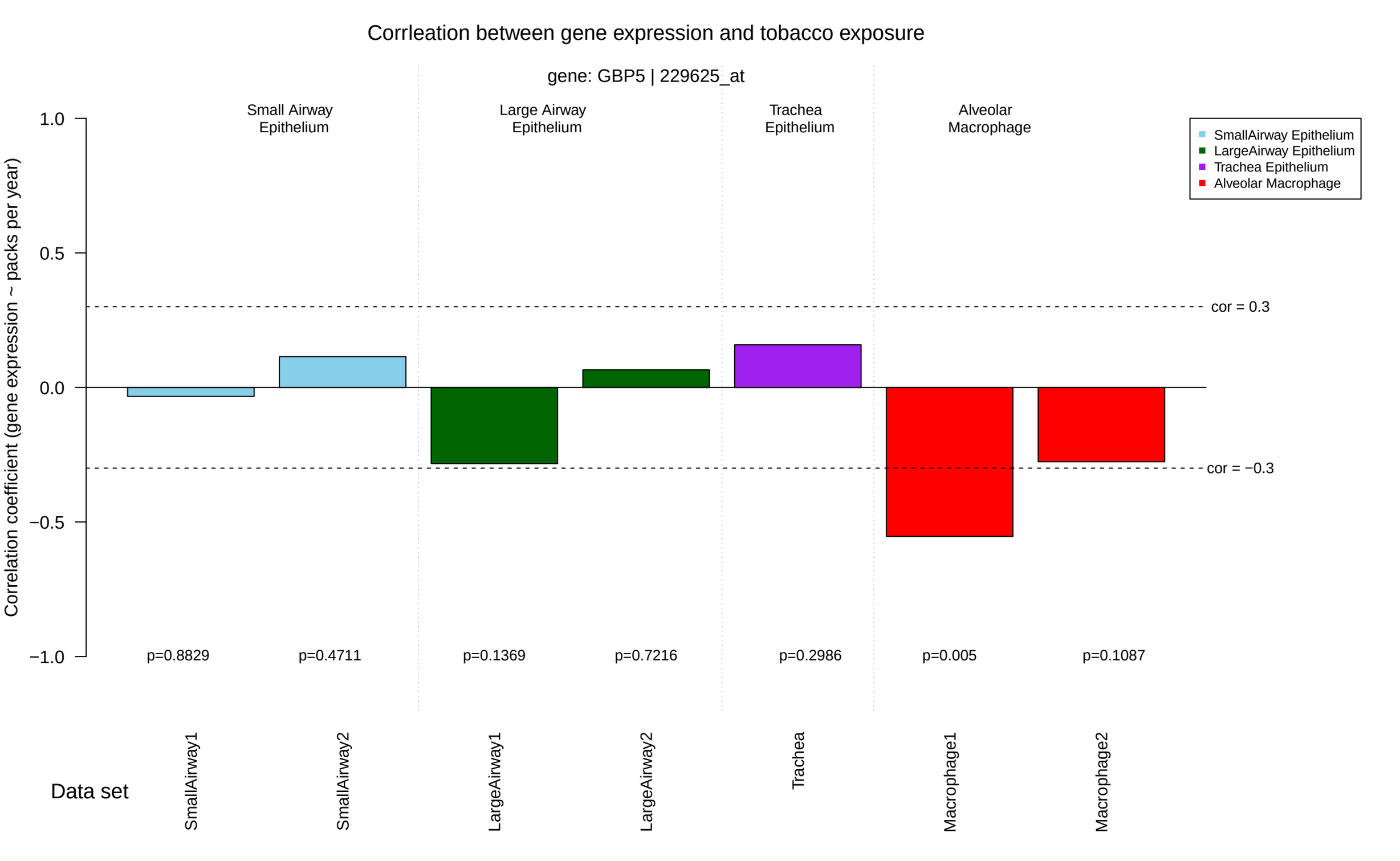

set: LargeAirway1 (p=0.5566)
GBP5 | 229625_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5145)
GBP5 | 229625_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3885)
GBP5 | 229625_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0023)
GBP5 | 229625_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0154)
GBP5 | 229625_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GBP5 | 229625_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8829
p=0.4711
p=0.1369
p=0.7216
p=0.2986
p=0.005
p=0.1087
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2